

Network Analysis

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SUMMARY:

Purpose: See the full attack picture across systems (C2, lateral movement, exfiltration).

When: After host analysis shows suspicious traffic or when scoping multiple systems.

Core Approach: Identify who talked to whom, how often, and why.

Key Steps:

1. Spot suspicious communication (unknown domains, high volume, strange ports).
2. Find infected hosts — filter by source IP in **Kibana**.
3. Track lateral movement (SMB port 445).
4. Investigate C2/beaconing (regular connections to same IP/port).
5. Detect scanning (many SYN without handshake).
6. Check exfiltration (DNS tunneling — high-entropy queries).
7. Extract IOCs (IPs, domains, techniques).

Main Tools: **Kibana** (Elasticsearch), **Zeek** (structured logs from PCAP), **tcpdump**. **Why it matters:** Reveals east-west traffic and the complete timeline that individual hosts miss.

When to perform: After host analysis reveals suspicious traffic, or when you need visibility across the environment. **Why:** Shows the full attack picture — C2, lateral movement, scanning, exfiltration — that individual hosts may miss. Builds an end-to-end timeline.

Main Tools:

- **Kibana** (with Elasticsearch): Fast log searching and filtering.
- **Zeek**: Turns PCAPs into structured logs (conn.log, http.log, dns.log, etc.).
- **tcpdump**: Raw packet capture and inspection.

Investigation Workflow (Step-by-Step Order)

1. Identify suspicious communication.
2. Find infected host(s).
3. Track lateral movement.
4. Investigate C2 and beaconing.
5. Detect scanning/enumeration.
6. Check critical servers and exfiltration.
7. Extract IOCs.

1. Identify Suspicious Communication

- **Tool:** **Kibana** Filter by unknown domains, strange ports, or high volume. **Why:** Spots anomalies like outbound traffic to rare domains.

2. Identify Infected Host

- **Tool:** **Kibana** Filter: source.ip == <suspect IP> **Why:** Narrows down which internal machine is communicating externally.

3. Track Lateral Movement

- **Tool:** **tcpdump** or **Kibana** (port 445/SMB) Example:
Bash
`tcpdump -nn -r incident.pcap host <IP>`
Why: Reveals internal SMB traffic used for file sharing or remote execution (common in ransomware spread).

4. Investigate C2 & Beaconing

- **Tool:** **Kibana** Filter for repeated connections to same IP/port (often 443). **Why:** Beaconing = malware calling home at regular intervals with consistent packet sizes.
- **Tool:** **Zeek** (cat conn.log or http.log) **Why:** Provides structured data on connections and downloads.

5. Detect Scanning

- **Tool:** **Kibana** or **tcpdump** Look for many SYN packets without full handshake (SYN scan). **Why:** Attackers scan for open ports or vulnerable systems.

6. Investigate Exfiltration & Critical Servers

- **Tool:** **Zeek** (dns.log + entropy scripts) High-entropy DNS queries indicate tunneling. **Why:** DNS can exfiltrate data covertly (high volume or encoded subdomains).

Mindset: Ask — Who talked to whom? How often? Why that port? Combine with host data for complete visibility. Always examine internal (east-west) traffic.

Goal: Build full attack timeline and identify all compromised assets + IOCs (IPs, domains, techniques).